

Analytics and Planning on Fiori

System design, and how it compares with SAP Analytics Cloud

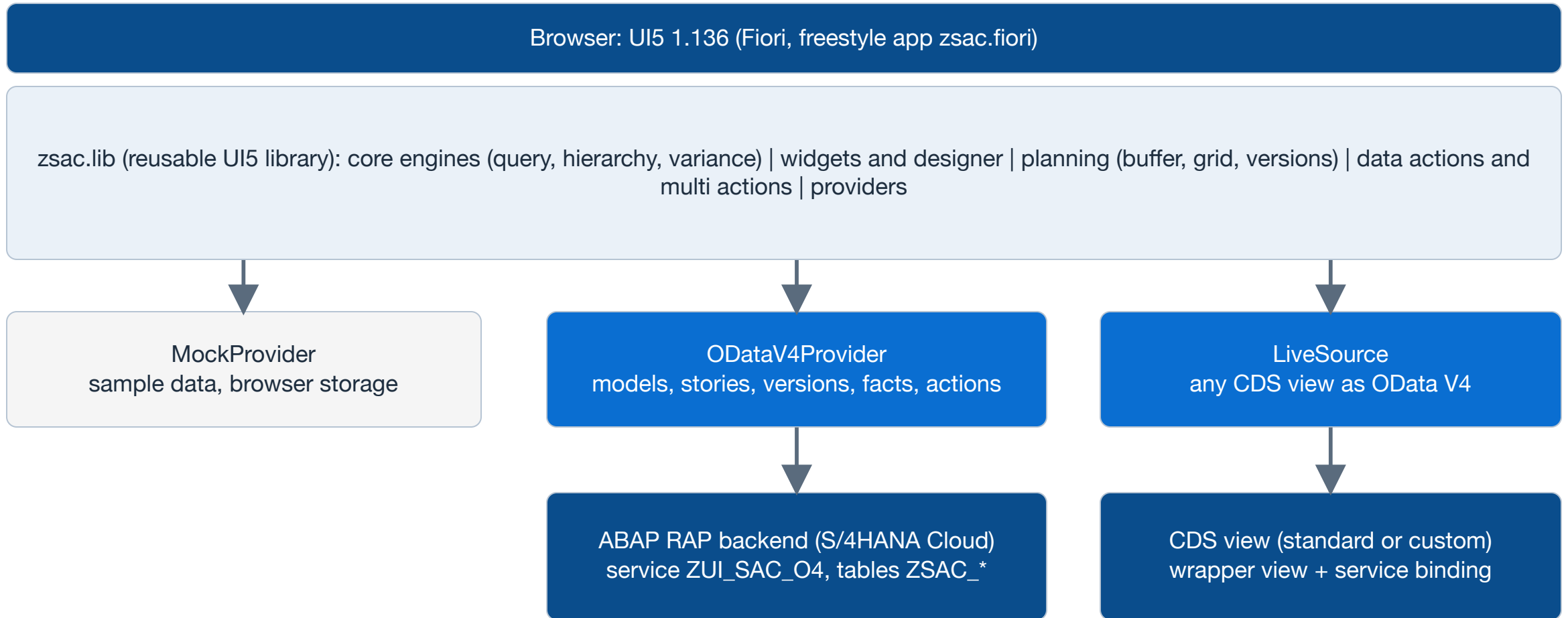
Why this was built

- Planning and analysis on S/4HANA data today needs a second product (SAC) with its own license, connections, transports and security.
- Question: how much of the SAC working model can run inside the ERP landscape, on the ERP's own authorizations?
- Answer so far: stories, analysis, models, planning with versions, data actions, multi actions, CDS data sources, variance analysis.
- Scope of this deck: the system design, and an honest comparison with the real SAC (advantages and gaps).

What it does (current release)

Area	Capability
Stories	Pages, grid designer, charts, KPI, tables, input controls, planning table, action triggers, variance widget
Models	SAC style Modeller: measures, dimensions, hierarchies, Date hierarchy, CSV import, CDS source
Planning	Editable table, spreading, formulas, copy and paste, versions, publish, history, comments on cells
Data actions	Designer, parameters, copy, allocation, scale, delete, embedded action; validate, trace, run history
Multi actions	Parameters, data action, publish, version, lock, CSV import, forecast, API, PaPM hook, comments, source import
CDS sources	Standard or custom CDS view as live (read only) or import source
Variance	Why did a measure change between versions or months, by every dimension, with drill-down

System design: the big picture



Pages and widgets only know the DataProvider contract. The browser works on facts; aggregation is one shared engine, so sample data and backend behave the same.

Design principles

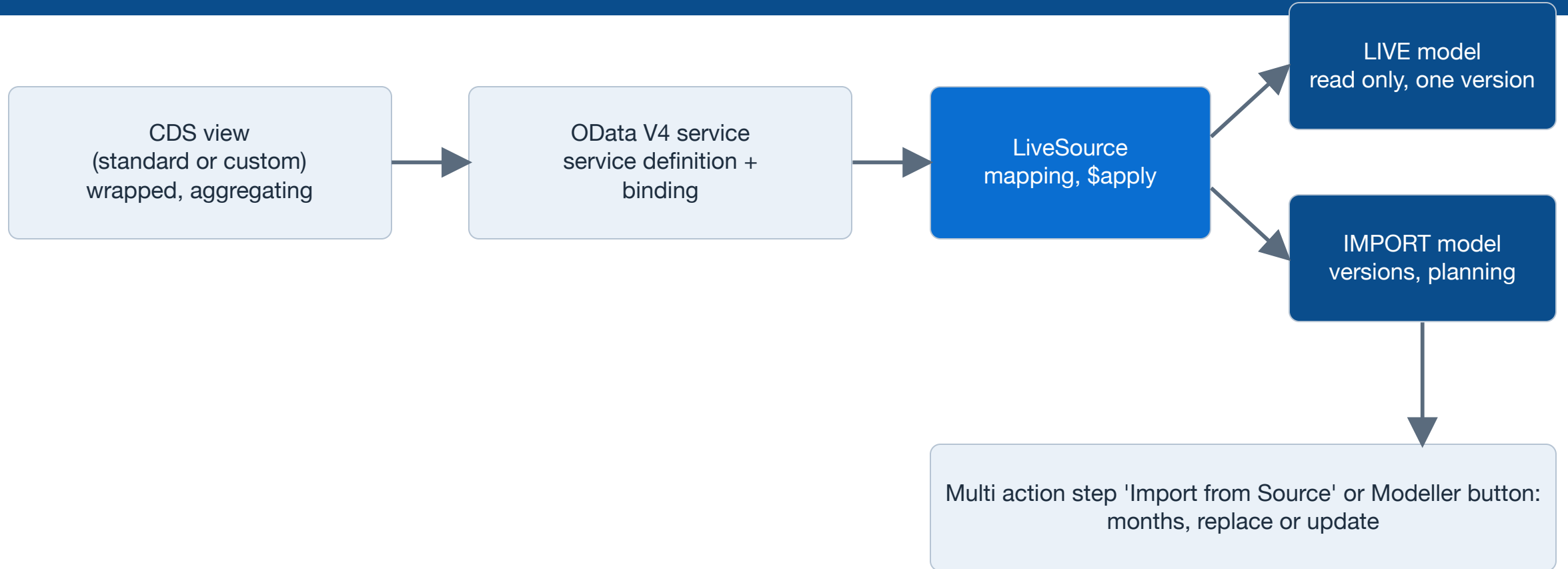
How it is built

- Modular: library `zsac.lib` is usable by any Fiori app; widgets and data sources are plug-ins registered in one place
- Metadata is data: models, stories, actions and files are rows edited by the same UI that shows them
- Generic fact table: version, period, measure and five dimension slots, so any model fits one RAP schema
- Provider contract: mock for development and tests, OData V4 for the backend, CDS for external data
- Generated backend: one description produces CDS, behavior and `abapGit` files; handlers by hand

Consequences

- Every feature can be tried and tested without a backend (77 unit tests)
- Backend stays small: tables, business objects, two engines, a seed class
- Calculations (actions, forecast, variance) run in the browser: easy to change, bounded by browser memory
- The ERP stays the system of record for authorization and plan data

Data sources: live and import

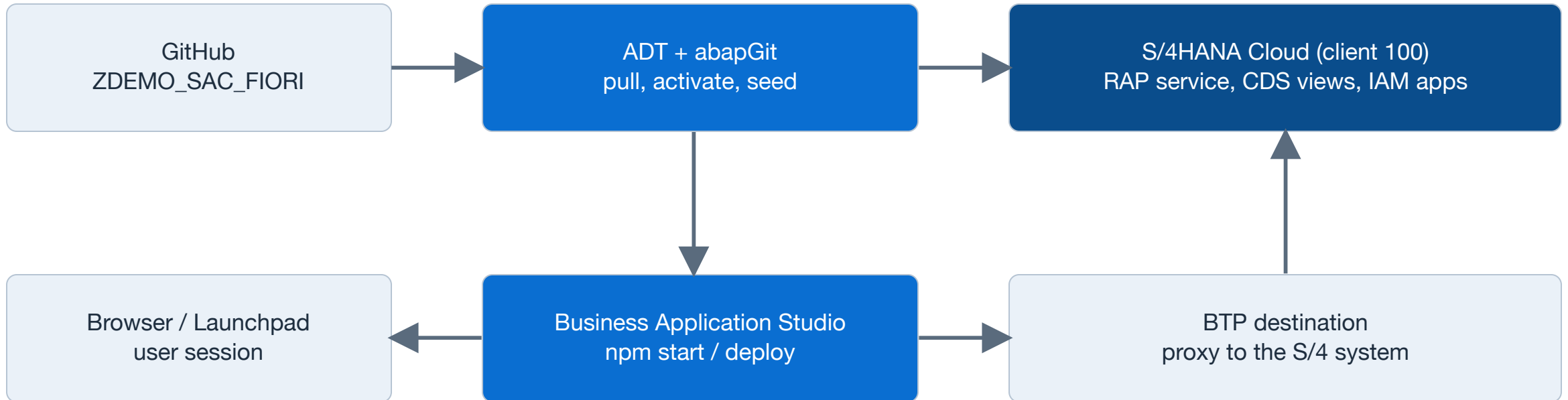


The service aggregates (groupby, aggregate); if it cannot, the app reads the rows and aggregates itself. Access control of the CDS view applies to the user who reads. Tested on a G/L view with 1,092 rows, two currencies.

Backend (ABAP RAP) at a glance

Part	Design
Tables	ZSAC_FILE, STORY, WIDGET, MODEL, DIM, MEASURE, VERSION, FACT, DATAACT, DASTEP, MULTIACT, MASTER, RUN, COMMENT, CALTASK
Business objects	Managed RAP, strict(2), no draft; compositions Story-Widget, Model-Dim/Measure, Action-Step; one request writes root and children
Actions	Version: create private, publish, revert. Fact: bulk write and delete
Rules on the server	Locked versions are not written; model structure checked on save; keys given by the caller
Generated	sap_spec.py to CDS, behavior, projections, service definition, abapGit metadata
Service	ZUI_SAC_O4 (OData V4 UI); UI5 app deployed as ZSAC_FIORI; launchpad descriptor after deployment

Landscape and delivery



Same code for mock, BAS test and deployment. A relative service URL lets the proxy or the launchpad add host and login. Where abapGit is not available the ABAP is entered by hand (a short list of changes).

Security and operations

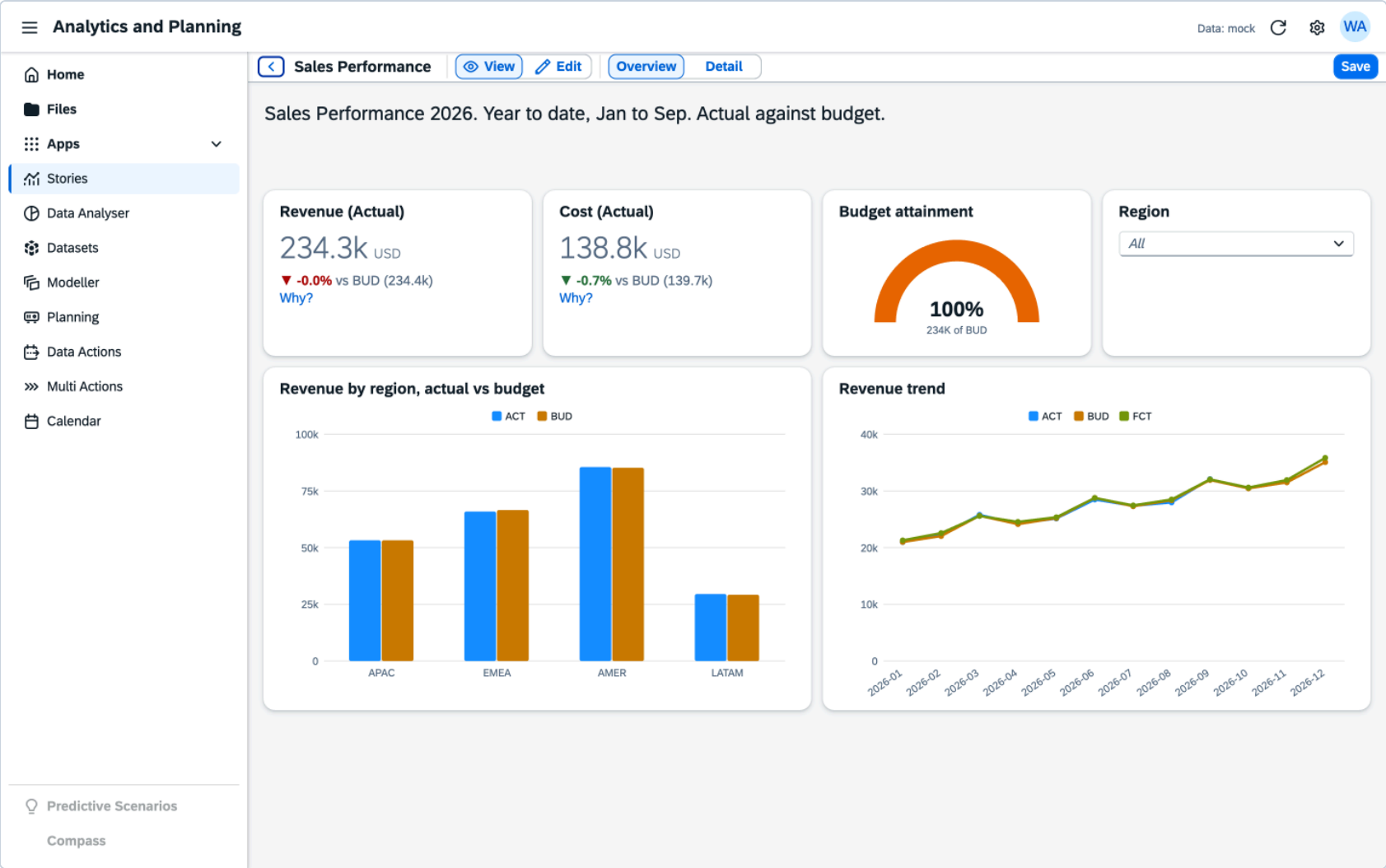
In place

- Calls run under the user's own session; CDS access control applies to live and import reads
- Custom services need an IAM app, business catalog and role (S/4HANA Cloud public edition)
- Locked versions refuse writes on the server and in every engine
- Every action run is recorded (status, counts, duration, user, parameters)
- Data audit and comments carry user and time from the backend

Not in place yet

- No owner or sharing rules for stories and models (open to all users of the service)
- No row level security on plan data (only on CDS source reads)
- No server job: actions run in the browser of the user
- No transport of content except through abapGit and the seed
- API step runs from the browser: no stored credentials, CORS applies

Stories: charts, KPIs, input controls



Story viewer on the sample data

Planning: edit, spread, comment, publish

Analytics and Planning

HomeFilesAppsStoriesData AnalyserDatasetsModellerPlanningData ActionsMulti ActionsCalendar

Predictive ScenariosCompass

Sales PlanningViewEditPlan

334.0K USD
▲ +1.0% vs BUD (331.4k)
Why?

Copies actuals into the forecast, then adds 5% for Q3
Regions (empty = all)
All
Q3 uplift factor

Save

Revenue forecast by region and month

Region	▼2026	►Q1 2026	►Q2 2026	►Q3 2026	▼Q4 2026	Oct 2026	Nov 2026	Dec 2026
▼ WORLD	334,601	69,482	78,715	88,009	98,395	30,611	31,932	35,852
▼ AMERICAS	163,439	33,820	38,469	42,933	48,217	14,979	15,831	17,406
AMER	121,515	25,171	28,618	31,844	35,882	11,142	11,745	12,995
LATAM	41,924	8,649	9,851	11,089	12,335	3,837	4,086	4,412
▼ EASTERN	171,163	35,662	40,246	45,076	50,179	15,632	16,102	18,445
APAC	75,977	15,693	17,888	19,975	22,420	6,927	7,317	8,176
EMEA	95,186	19,970	22,358	25,100	27,759	8,705	8,784	10,269

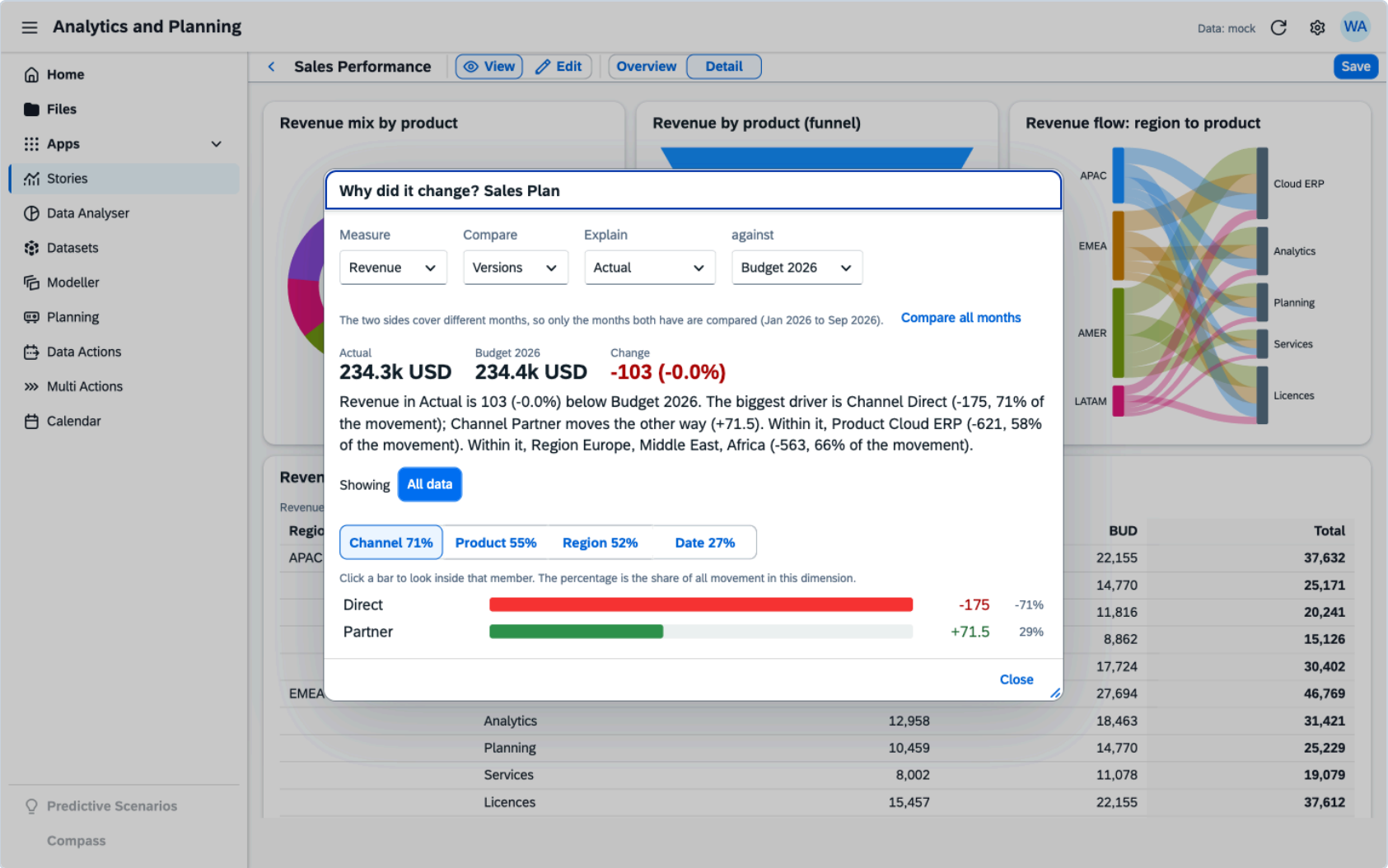
Editable planning table; the orange corner marks a cell with a comment

Data actions: designer with step flow

The screenshot displays the 'Analytics and Planning' interface. On the left is a sidebar with navigation options: Home, Files, Apps, Stories, Data Analyser, Datasets, Modeller, Planning, Data Actions (highlighted), Multi Actions, and Calendar. Below this is a 'Predictive Scenarios' section with a 'Compass' icon. The main area is titled 'Forecast Q4 from run-rate' and contains a step flow diagram with two steps: '1. Copy actuals to forecast' (labeled 'Copy') and '2. Uplift Q3' (labeled 'Scale'). To the right of the flow is the 'Copy step' editor. It includes a 'Name' field with the value 'Copy actuals to forecast', a 'Description' field with 'Overwrites the forecast of the chosen regions', and a 'Step Active' toggle set to 'ON'. The 'Filter (which facts the step works on)' section shows 'Version' set to 'ACT - Actual' and 'Region' set to '@Region (parameter)'. The 'Copy rules (from → to, per dimension)' section shows a rule from 'Version' to 'FCT'. Below this is an 'Aggregate to' section with an 'Add aggregation' button. The 'Write mode' is set to 'Overwrite', and the 'Copy factor' is set to '1'. A toolbar at the top right of the editor contains icons for 'Settings', 'Parameters (2)', 'Add Step', 'Validate', 'Trace', and 'Run'. The top right corner of the interface shows 'Data: mock', a refresh icon, a settings icon, and a user profile icon labeled 'WA'.

Step flow on the left, step editor on the right; validate, trace and run on the toolbar

Variance explainer: why did it change?



Headline, named drivers, ranked dimensions, bars to drill into

Tested on the real system (S/4HANA Cloud, OData V4)

Feature	Result
Backend reads, live CDS model, story with filters	Works
Save model (deep insert), versions, import of 1,092 values	Works
Lock and unlock (PATCH), data kept	Works
Data action with parameters: dry run and run (329 facts)	Works, result checked
Multi action: data action, lock, forecast, comments, source import, CSV import	Works (8 steps, 85 s)
Planning edit and publish, cell comments, variance explainer	Works
Create private version (server action)	Fails until two ABAP lines are changed by hand
API and PaPM steps	As designed: target 401 reported, PaPM not connected

Versus real SAC: capability

Capability	This app	SAC
Stories, analysis, models, hierarchies	Core set	Complete, mature
Planning with versions, publish, history	Yes	Yes, plus workflows and calendar approvals
Data actions	Copy, allocate, scale, delete, embed, parameters	Plus advanced formulas, currency conversion, cross-model
Multi actions	10 step types	Similar, more connectors
Predictive / ML, Smart Insights	Statistical forecast, variance explainer	Smart Predict, Smart Insights, Just Ask
Scripting, custom widgets	Widget registry in code	Analytics Designer scripting
Mobile, Excel add-in, subscriptions	No	Yes
Data connections	CDS via OData V4	Live and import, many sources incl. BW, S/4
Scale	Browser memory bound	Server side, large volumes

Advantages over using real SAC

Pros

- No extra product, license or tenant: runs in the S/4HANA landscape and the Fiori launchpad
- One security model: the user's CDS access control, no gateway, SSO setup or connection admin
- Plan data lives in the ERP (RAP tables): no replication, same transport and backup path, close to the source
- Open and changeable: every step, widget and rule can be adapted in code in days
- Tailored features: CDS live and import, variance explainer, multi action steps built for this use
- Modular library reusable in other Fiori apps; sample mode for training and tests

Cons: gaps compared with SAC

- Far narrower scope: no Smart Predict, Smart Insights, Just Ask, mobile app, Excel add-in, subscriptions
- Calculations run in the browser: large models and long multi actions are slow (85 s for 8 steps on 1,092 rows)
- Owned code: we maintain it, no SAP support or release roadmap, UI5 and RAP upgrades are our work
- Security and governance are basic: no sharing rules, no row level plan security, no content transport
- Planning depth is smaller: no workflows or approvals, no advanced formulas, no currency conversion
- Only validated on one system; ABAP needed hand fixes on a system without abapGit

When to use which

Use this app when

- The data is in S/4HANA CDS views and the planning is modest: versions, allocations, forecasts
- A second product or license is not wanted, or data may not leave the ERP
- Users are already in Fiori and need a few tailored dashboards and plan tables
- A fast, customizable pilot is wanted before a larger tool decision

Use real SAC when

- Enterprise planning: workflows, approvals, many users, large data volumes
- Many source systems (BW, non-SAP) are combined
- Advanced analytics, predictive, natural language, mobile and Excel are required
- SAP support, content governance and a product roadmap are required

Status and next steps

- Status: complete feature set on the sample data; verified on a real system except one ABAP fix (private versions).
- 1. Apply the ABAP fix on the system without abapGit, then test publish, revert, delete of private versions.
- 2. Decide the owner and sharing model (who sees which story and model) and add authorization.
- 3. Measure with realistic volume (CDS rows, facts per version) and decide whether actions need a server job.
- 4. Pilot with one finance team on the G/L model: live analysis first, then import and a budget version.
- 5. Decide against SAC with the pilot results: which gaps matter to the users.